

Question: Define sampling distribution, linear orthogonal transformation.

Answer: Sampling distribution: Sampling distribution is the probability distribution of a statistic, derived from a random sample, drawn from some specified parent population. For example χ^2 , t and F -distributions are sampling distribution.

Linear Orthogonal Transformation: Suppose $x_1, x_2, \dots, x_n$ is a random sample of n observations. Consider two linear functions

$$l_1 = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$$

$$\text{and } l_2 = b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

Here l_1 and l_2 are said to be uncorrelated or independent iff $a_1 b_1 + a_2 b_2 + \dots + a_n b_n = 0$.

Two uncorrelated linear functions are said to be orthogonal. Consider the following linear transformations

$$\left. \begin{aligned} y_1 &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ y_2 &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ &\vdots \\ y_n &= a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n \end{aligned} \right\} \rightarrow \textcircled{1}$$

The transformation given by $\textcircled{1}$ may be written in matrix notation as $y = AX$,

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

A linear function $y = AX$ is said to be linear orthogonal transformation from x to y if the transformation matrix A is an orthogonal matrix.

Question: If y_i ($i=1,2,\dots,n$) are connected by linear orthogonal transformation $y_i = \sum_{j=1}^n a_{ij} x_j$, $i=1,2,\dots,n$, then show that $\sum_{i=1}^n y_i^2 = \sum_{i=1}^n x_i^2$ and the jacobian of the transformation is ± 1 .

Proof: We have $\sum_{i=1}^n y_i^2 = (y_1 \ y_2 \ \dots \ y_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$

$$= y'y$$

$$= (AX)'(AX)$$

$$= x'A'AX$$

$$= x'x = (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$= \sum_{i=1}^n x_i^2$$

Hence $\sum_{i=1}^n y_i^2 = \sum_{i=1}^n x_i^2$ (Proved).

Since $y = AX$

Now premultiplying both sides by A' we have

$$A'y = A'AX$$

$$\Rightarrow x = A'y$$

$$\Rightarrow |J| = \left| \frac{dx}{dy} \right| = |A'|$$

$$= \pm 1 \text{ (proved)}$$

Since $AA' = I$

$$\Rightarrow |AA'| = |I|$$

$$\Rightarrow |A||A'| = 1$$

$$\Rightarrow |A'|^2 = 1$$

$$\Rightarrow |A'| = \pm 1$$

Question: State and prove Fisher's Lemma.

Answer:

Statement: If $x_1, x_2, \dots, x_n$ is a random sample from a normal distribution with mean zero and variance σ^2 and if A is an orthogonal matrix producing the new variables $y_i = \sum_{j=1}^n a_{ij} x_j$, then $y_1, y_2, \dots, y_n$ is also a random sample from the normal distribution with mean zero and variance σ^2 .

$$\left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n$$

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Proof: The joint pdf of $x_1, x_2, \dots, x_n$ is

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f(x_1) f(x_2) \dots f(x_n) \\ &= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x_1^2}{2\sigma^2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x_2^2}{2\sigma^2}} \dots \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x_n^2}{2\sigma^2}} \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n e^{-\sum_{i=1}^n \frac{x_i^2}{2\sigma^2}} \quad ; \quad -\infty < x_i < \infty \\ &\quad i=1, 2, \dots, n \end{aligned}$$

Since $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ are connected by linear orthogonal transformation then $\sum_{i=1}^n y_i^2 = \sum_{i=1}^n x_i^2$ and $|J| = \pm 1$.

Then the joint pdf of $y_1, y_2, \dots, y_n$ is

$$\begin{aligned} f(y_1, y_2, \dots, y_n) &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n e^{-\sum_{i=1}^n \frac{y_i^2}{2\sigma^2}} |J| \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n e^{-\sum_{i=1}^n \frac{y_i^2}{2\sigma^2}} (\pm 1) \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n e^{-\sum_{i=1}^n \frac{y_i^2}{2\sigma^2}} \quad ; \quad -\infty < y_i < \infty \\ &\quad i=1, 2, \dots, n \end{aligned}$$

Hence $y_1, y_2, \dots, y_n$ be a random sample from the normal distribution with mean zero and variance σ^2 . (Proved)

Degrees of Freedom: Degrees of freedom is the number of values in a sample we can specify freely. for example, let the mean of a sample of 4 observations is $\bar{x} = 15$ then take $x_1 = 50, x_2 = 20, x_3 = -35$ and then $x_4 = 4 \times 15 - (50 + 20 - 35) = 25$. Then the degrees of freedom is $4 - 1 = 3$.

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Question: χ^2 -variate/ χ^2 -statistic. Derive the χ^2 -distribution.

Answer:

χ^2 -variate/ χ^2 -statistic: The square of a standard normal variate is known as a chi-square (χ^2) variate with 1 d.f.

Thus if $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ and $Z^2 = \left(\frac{X-\mu}{\sigma}\right)^2$ is a χ^2 -variate with 1 d.f.

In general, if $X_i (i=1,2,\dots,n)$ are n independent normal variates with mean μ_i and variance $\sigma_i^2 (i=1,2,\dots,n)$, then $\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2$ is a χ^2 -variate/ χ^2 -statistic with n degrees of freedom (d.f.).

Derivation of the χ^2 -distribution: If $X_i (i=1,2,\dots,n)$ are independent $N(\mu_i, \sigma_i^2)$ variates then the quantity $\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2$ is distributed as χ^2 with n d.f.

Proof: Let $Z_i = \frac{X_i - \mu_i}{\sigma_i}, i=1,2,\dots,n$

Since $X_i \sim N(\mu_i, \sigma_i^2)$ then $Z_i \sim N(0,1)$

Now the moment generating function of χ^2 is given by

$$M_{\chi^2}(t) = M_{\sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2}(t) = M_{\sum_{i=1}^n Z_i^2}(t) = \prod_{i=1}^n M_{Z_i^2}(t)$$

$$\text{But } M_{Z_i^2}(t) = E(e^{tZ_i^2}) = \int_{-\infty}^{\infty} e^{tZ_i^2} f(Z_i) dZ_i$$

$$= \int_{-\infty}^{\infty} e^{tZ_i^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z_i^2} dZ_i$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{1-2t}{2}\right)Z_i^2} dZ_i$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1-2t}{2}Z_i^2} Z_i^{\frac{1}{2}-1} dZ_i \quad \left| \text{Since } f \text{ is even function} \right.$$

$$= \frac{2}{\sqrt{2\pi}} \frac{\left(\frac{1-2t}{2}\right)^{-\frac{1}{2}} \Gamma\left(\frac{1}{2}\right)}{2} \quad \left| \because \int_0^{\infty} e^{-ax^2} x^{2n-1} dx = \frac{a^{-n} \Gamma(n)}{2} \right.$$

$$= \frac{1}{(1-2t)^{1/2}}$$

$$\left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \right]$$

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$$\therefore M_{x^2}(t) = \prod_{i=1}^n M_{x_i^2}(t) = \prod_{i=1}^n \frac{1}{(1-2t)^{1/2}} = \frac{1}{2(1-2t)^{n/2}}$$

$$= \left(\frac{1/2}{1/2-t} \right)^{n/2}$$

which is the moment generating function of a gamma variate with parameter $\lambda = \frac{1}{2}$ and $r = n/2$, [since $G(\lambda, r) = \frac{\lambda^r}{\Gamma(r)} e^{-\lambda x} x^{r-1}, 0 < x < \infty$].

Therefore the pdf of x^2 is

$$dF(x^2) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} e^{-\frac{1}{2}x^2} (x^2)^{\frac{n}{2}-1} dx^2 = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(n/2)} dx^2; 0 \leq x^2 < \infty$$

$$\text{or, } f(x^2) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} e^{-\frac{1}{2}x^2} (x^2)^{\frac{n}{2}-1} = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(n/2)}; 0 \leq x^2 < \infty$$

which is the pdf of a χ^2 -distribution with n d.f.

Question: Find the mean, variance and mode of χ^2 -distribution with n d.f.

Answer: We know that the pdf of χ^2 -distribution with n d.f. is

$$f(x^2) = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(n/2)}; 0 \leq x^2 < \infty$$

Mean of x^2 :

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x^2) dx^2$$

$$= \int_0^{\infty} x^2 \cdot \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(n/2)} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^{\infty} e^{-x^2/2} (x^2)^{(\frac{n}{2}+1)-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \cdot \frac{1}{2} \left(\frac{1}{2} \right)^{-(\frac{n}{2}+1)} \Gamma\left(\frac{n}{2}+1\right) \quad \left| \because \int_0^{\infty} e^{-ax} x^{n-1} dx = a^{-n} \Gamma(n) \right.$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \cdot 2^{\frac{n}{2}+1} \cdot \frac{n}{2} \Gamma\left(\frac{n}{2}\right)$$

$$= n$$

Variance of χ^2 :

We know that $V(\chi^2) = E\{(\chi^2)^2\} - \{E(\chi^2)\}^2$

But $E\{(\chi^2)^2\} = \int_{\chi^2} (\chi^2)^2 f(\chi^2) d\chi^2$

$$= \int_0^{\infty} (\chi^2)^2 \frac{e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma_{n/2}} d\chi^2$$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} \int_0^{\infty} e^{-\chi^2/2} (\chi^2)^{(\frac{n}{2}+2)-1} d\chi^2$$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} \left(\frac{1}{2}\right)^{-(\frac{n}{2}+2)} \Gamma_{\frac{n}{2}+2}$$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} 2^{\frac{n}{2}+2} \left(\frac{n}{2}+1\right) \left(\frac{n}{2}\right) \Gamma_{\frac{n}{2}}$$

$$= n(n+2)$$

$$\therefore V(\chi^2) = n(n+2) - n^2 = 2n$$

Mode of χ^2 :

Here $f(\chi^2) = \frac{e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma_{n/2}}, 0 \leq \chi^2 < \infty$

Now $f'(\chi^2) = \frac{\partial f(\chi^2)}{\partial \chi^2} = \frac{1}{2^{n/2} \Gamma_{n/2}} \left[\left(-\frac{1}{2}\right) e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1} + e^{-\chi^2/2} \left(\frac{n}{2}-1\right) (\chi^2)^{\frac{n}{2}-2} \right]$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1} \left[\left(\frac{n}{2}-1\right) \frac{1}{\chi^2} - \frac{1}{2} \right] = \left[\left(\frac{n}{2}-1\right) \frac{1}{\chi^2} - \frac{1}{2} \right] f(\chi^2)$$

$$\Rightarrow \left(\frac{n}{2}-1\right) \frac{1}{\chi^2} - \frac{1}{2} = 0 \quad \left| \quad \text{Since } \frac{1}{2^{n/2} \Gamma_{n/2}} e^{-\chi^2/2} (\chi^2)^{\frac{n}{2}-1} \neq 0 \right.$$

$$\Rightarrow \chi^2 = n-2$$

We can easily show that $f''(\chi^2) < 0$ when $\chi^2 = n-2$.
Therefore the mode of χ^2 -distribution is $n-2$. ✓

Question: Find the moment generating function of a χ^2 -distribution with n d.f. Also find the mean, variance, skewness, kurtosis and comment.

Answer: We know that the pdf of χ^2 -distribution with n d.f. is given by

$$f(x^2) = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma_{n/2}}; 0 \leq x^2 < \infty$$

The mgf of χ^2 -distribution is

$$M_{x^2}(t) = E(e^{tx^2}) = \int_0^\infty e^{tx^2} f(x^2) dx^2$$

$$= \int_0^\infty e^{tx^2} \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma_{n/2}} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} \int_0^\infty e^{-(\frac{1}{2}-t)x^2} (x^2)^{\frac{n}{2}-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma_{n/2}} \left(\frac{1-2t}{2} \right)^{-\frac{n}{2}} \Gamma_{n/2} \quad \because \int_0^\infty e^{-ax} x^{n-1} dx = a^{-n} \Gamma_n$$

$$= (1-2t)^{-n/2} \text{ which is the mgf of } \chi^2\text{-distribution.}$$

$$\text{Mean} = \mu_1' = \frac{d}{dt} M_{x^2}(t) \big|_{t=0}$$

$$= \left(-\frac{n}{2} \right) (1-2t)^{-\frac{n}{2}-1} (-2) \big|_{t=0}$$

$$= n (1-2t)^{-\frac{n}{2}-1} \big|_{t=0} = n$$

$$\mu_2' = \frac{d^2}{dt^2} M_{x^2}(t) \big|_{t=0} = \frac{d}{dt} \left[n(1-2t)^{-\frac{n}{2}-1} \right] \big|_{t=0}$$

$$= n(n+2) (1-2t)^{-\frac{n}{2}-2} \big|_{t=0}$$

$$= n^2 + 2n$$

$$\therefore \text{Variance} = \sigma^2 = \mu_2 = \mu_2' - \mu_1'^2$$

$$= n^2 + 2n - n^2 = 2n$$

$$\begin{aligned}\mu_3' &= \frac{d^3}{dt^3} M_{X^2}(t) \big|_{t=0} = \frac{d}{dt} \left[n(n+2)(1-2t)^{-\frac{n}{2}-2} \right] \big|_{t=0} \\ &= n(n+2)(n+4)(1-2t)^{-\frac{n}{2}-3} \big|_{t=0} \\ &= n(n+2)(n+4)\end{aligned}$$

$$\therefore \mu_3 = \mu_3' - 3\mu_2' \mu_1' + 2\mu_1'^3$$

$$\begin{aligned}&= n(n+2)(n+4) - 3n(n+2)n + 2n^3 \\ &= n(n+2)(n+4) - n^2(n+2) - 2n^2(n+2) + 2n^3 \\ &= n(n+2)(n+4-n) - 2n^2(n+2-n) \\ &= 4n(n+2) - 4n^2 \\ &= 4n(n+2-n) = 8n\end{aligned}$$

$$\begin{aligned}\mu_4' &= \frac{d^4}{dt^4} M_{X^2}(t) \big|_{t=0} = \frac{d}{dt} \left[n(n+2)(n+4)(1-2t)^{-\frac{n}{2}-3} \right] \big|_{t=0} \\ &= n(n+2)(n+4)(n+6)(1-2t)^{-\frac{n}{2}-4} \big|_{t=0} \\ &= n(n+2)(n+4)(n+6)\end{aligned}$$

$$\therefore \mu_4 = \mu_4' - 4\mu_3' \mu_1' + 6\mu_2' \mu_1'^2 - 3\mu_1'^4$$

$$\begin{aligned}&= n(n+2)(n+4)(n+6) - 4n(n+2)(n+4)n + 6n(n+2)n^2 - 3n^4 \\ &= n(n+2)(n+4)(n+6) - 4n^2(n+2)(n+4) + 6n^3(n+2) - 3n^4 \\ &= n(n+2)(n+4)(n+6-n) - 3n^2(n+2)(n+4-n) + 3n^3(n+2-n) \\ &= 6n(n+2)(n+4) - 12n^2(n+2) + 6n^3 \\ &= 6n(n+2)(n+4-n) - 6n^2(n+2-n) \\ &= 24n(n+2) - 12n^2 \\ &= 24n^2 + 48n - 12n^2 \\ &= 12n^2 + 48n\end{aligned}$$

Skewness, $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(8n)^2}{(2n)^3} = \frac{8}{n}$

∴ Coefficient of skewness, $\sqrt{\beta_1} = \sqrt{\frac{8}{n}}$

Kurtosis, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{12n^2 + 48n}{4n^2} = 3 + \frac{12}{n}$

Comment: Since $\mu_1' = n$ and $\mu_2 = 2n$ then the mean of χ^2 -distribution is equal to its d.f. and the variance of χ^2 -distribution is double its d.f. Also since $\sqrt{\beta_1} = \sqrt{\frac{8}{n}}$ and $\beta_2 = 3 + \frac{12}{n}$ then the χ^2 -distribution is positively skewed and leptokurtic.

Question: state and prove the additive property of χ^2 -distribution.

Answer: Statement: If $\chi_1^2, \chi_2^2, \dots, \chi_k^2$ are k independent χ^2 variates with corresponding d.f. $n_1, n_2, \dots, n_k$. Then $\chi^2 = \chi_1^2 + \chi_2^2 + \dots + \chi_k^2$ follows χ^2 -distribution with $n = n_1 + n_2 + \dots + n_k$ d.f.

Proof: We know that

$$\begin{aligned} M_{\chi^2}(t) &= M_{\chi_1^2 + \chi_2^2 + \dots + \chi_k^2}(t) \\ &= M_{\chi_1^2}(t) \cdot M_{\chi_2^2}(t) \cdot \dots \cdot M_{\chi_k^2}(t) \\ &= (1-2t)^{-n_1/2} (1-2t)^{-n_2/2} \dots (1-2t)^{-n_k/2} \\ &= (1-2t)^{-\frac{1}{2}(n_1 + n_2 + \dots + n_k)} \end{aligned}$$

Since the mgf of χ^2 with n d.f. is $(1-2t)^{-n/2}$

$$\Rightarrow M_{\chi^2}(t) = (1-2t)^{-\frac{1}{2}(n_1 + n_2 + \dots + n_k)}$$

which is the mgf of a χ^2 variate with $n = n_1 + n_2 + \dots + n_k$ d.f.

Hence $\chi^2 = \chi_1^2 + \chi_2^2 + \dots + \chi_k^2$ is a χ^2 -variate with $n = n_1 + n_2 + \dots + n_k$ degrees of freedom.

Question: Show that χ^2 -distribution approaches to normal distribution as $n \rightarrow \infty$. OR Show that χ^2 -distribution tends to normal distribution for large degrees of freedom.

Answer:

$$\text{Let } Z = \frac{\chi^2 - E(\chi^2)}{\sqrt{\text{Var}(\chi^2)}} = \frac{\chi^2 - n}{\sqrt{2n}} = \frac{\chi^2}{\sqrt{2n}} - \sqrt{\frac{n}{2}}$$

The mgf of standard χ^2 -variate Z is given by

$$M_Z(t) = E(e^{tZ}) = E\left[e^{\left(\frac{\chi^2}{\sqrt{2n}} - \sqrt{\frac{n}{2}}\right)t}\right]$$

$$= e^{-\sqrt{\frac{n}{2}}t} E\left(e^{\frac{\chi^2 t}{\sqrt{2n}}}\right)$$

$$= e^{-t\sqrt{\frac{n}{2}}} M_{\chi^2}\left(\frac{t}{\sqrt{2n}}\right) \quad \left| \text{Since } M_{a+bx}(t) = e^{at} \cdot M_x(bt) \right.$$

$$= e^{-t\sqrt{\frac{n}{2}}} \left(1 - \frac{2t}{\sqrt{2n}}\right)^{-n/2}$$

$$\Rightarrow M_Z(t) = e^{-t\sqrt{\frac{n}{2}}} \left(1 - t\sqrt{\frac{2}{n}}\right)^{-n/2}$$

Taking logarithm of both sides then we get

$$\log M_Z(t) = -t\sqrt{\frac{n}{2}} - \frac{n}{2} \log\left(1 - t\sqrt{\frac{2}{n}}\right)$$

$$= -t\sqrt{\frac{n}{2}} + \frac{n}{2} \left[t\sqrt{\frac{2}{n}} + \frac{2t^2}{2n} + \frac{t^3}{2} \left(\frac{2}{\sqrt{n}}\right)^3 + \dots \right]$$

$$= -t\sqrt{\frac{n}{2}} + t\sqrt{\frac{n}{2}} + \frac{1}{2}t^2 + O\left(\frac{1}{\sqrt{n}}\right)$$

$$= \frac{t^2}{2} + O\left(\frac{1}{\sqrt{n}}\right)$$

where $O\left(\frac{1}{\sqrt{n}}\right)$ are terms containing $\sqrt{n}$ and higher powers as n in the denominator.

$$\therefore \lim_{n \rightarrow \infty} \log M_Z(t) = \frac{t^2}{2} + 0 = \frac{t^2}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} M_Z(t) = e^{\frac{t^2}{2}}$$

which is the mgf of a standard normal variate. Thus as $n \rightarrow \infty$ a standard χ^2 -variate approaches as standard normal variate.

For large degrees of freedom χ^2 -distribution tends to normal distribution.

$$\begin{aligned} \log(1+x) &= x + \frac{x^2}{2} + \frac{x^3}{3} \\ x &= -t\sqrt{\frac{2}{n}} \\ &= -\frac{1}{2}t\sqrt{\frac{2}{n}} \\ &= -\frac{t}{\sqrt{n}} \end{aligned}$$

Question: Write down the properties and applications of χ^2 distⁿ.

Answer:

Properties of χ^2 distribution: The properties of χ^2 distribution are given below -

i) The pdf of χ^2 -distribution with n d.f is

$$f(x^2) = \frac{e^{-x^2/2} (x^2)^{\frac{n}{2}-1}}{2^{n/2} \Gamma(\frac{n}{2})}; 0 \leq x^2 < \infty$$

ii) The range of χ^2 -distribution is zero to infinity.

iii) The mean of χ^2 -distribution is equal to its d.f and the variance is equal to its twice d.f.

iv) This distribution is positively skewed and leptokurtic.

v) Its mode is $n-2$.

vi) χ^2 -distribution approaches normal distribution as $n \rightarrow \infty$.

vii) If $X \sim N(0,1)$ then $X^2 \sim \chi^2$ with 1 d.f.

Applications of χ^2 distribution: χ^2 -distribution has a large no. of applications in statistics, some of which are enumerated below -

i) To test if the hypothetical value of the population variance is $\sigma^2 = \sigma_0^2$ (say).

ii) To test the 'goodness of fit'.

iii) To test the independence of attributes.

iv) To test the homogeneity of independent estimates of the population variance.

v) To test the homogeneity of independent estimates of the population correlation co-efficient.

Theorem: If $X \sim N(0,1)$ then $X^2 \sim \chi^2$ -distribution with 1 d.f.

Proof: Since $X \sim N(0,1)$, then

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad ; -\infty < x < \infty$$

Now the mgf of X^2 is

$$\begin{aligned} M_{X^2}(t) &= E(e^{tx^2}) = \int_{-\infty}^{\infty} e^{tx^2} f(x) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{tx^2} e^{-x^2/2} dx = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{\frac{1-2t}{2}x^2} x^{2 \cdot \frac{1}{2} - 1} dx \\ &= \frac{2}{\sqrt{2\pi}} \frac{\left(\frac{1-2t}{2}\right)^{-1/2} \Gamma\left(\frac{1}{2}\right)}{2} = (1-2t)^{-1/2} \end{aligned}$$

which is the mgf of χ^2 -distribution with 1 d.f.

Hence if $X \sim N(0,1)$ then $X^2 \sim \chi^2$ -distribution with 1 d.f.

Theorem: If x has a density function $f(x) = e^{-x}$, $x > 0$ then $2x$ follows χ^2 -distribution with 2 d.f.

Proof:

The mgf of $2x$ is

$$\begin{aligned} M_{2x}(t) &= E(e^{2tx}) = \int_0^{\infty} e^{2tx} e^{-x} dx \\ &= \int_0^{\infty} e^{-(1-2t)x} x^{1-1} dx \\ &= (1-2t)^{-1} \Gamma(1) = (1-2t)^{-2/2} \end{aligned}$$

which is the mgf of χ^2 -distribution with 2 d.f.

Theorem: Fisher's results that, $\sqrt{2}X^2$ follows, approximately normal distribution with mean $\sqrt{2n-1}$ and variance unity.

Proof: Let $Z = \frac{X^2 - E(X^2)}{\sqrt{\text{Var}(X^2)}}$

$\Rightarrow \lim_{n \rightarrow \infty} M_2(t) = e^{t^2/2}$. Thus Z is an $N(0,1)$ variate as $n \rightarrow \infty$.

Now

$$\begin{aligned} P[\sqrt{2}X^2 - \sqrt{2n-1} \leq x] \\ &= P[\sqrt{2}X^2 \leq x + \sqrt{2n-1}] \\ &= P[2X^2 \leq x^2 + 2\sqrt{2n-1} \cdot x + 2n-1] \\ &= P[X^2 \leq \frac{x^2}{2} + x\sqrt{2n-1} + n - \frac{1}{2}] \\ &= P[X^2 \leq \frac{x^2}{2} + (n - \frac{1}{2}) + x\sqrt{2n-1}] \\ &= P[\frac{X^2}{\sqrt{2n}} \leq \frac{x^2}{2\sqrt{2n}} + \frac{n}{\sqrt{2n}} - \frac{1}{2\sqrt{2n}} + \frac{x\sqrt{2n-1}}{\sqrt{2n}}] \\ &= P[\frac{X^2}{\sqrt{2n}} - \frac{n}{\sqrt{2n}} \leq \frac{x^2}{2\sqrt{2n}} - \frac{1}{2\sqrt{2n}} + \frac{x\sqrt{2n-1}}{\sqrt{2n}}] \\ &= P[\frac{X^2 - n}{\sqrt{2n}} \leq \frac{x^2}{2\sqrt{2n}} - \frac{1}{2\sqrt{2n}} + x(\sqrt{1 - \frac{1}{2n}})] \\ &= P[\frac{X^2 - n}{\sqrt{2n}} \leq x] \text{ as } n \rightarrow \infty \\ &\text{which is normal cdf.} \end{aligned}$$

$\therefore F(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2n}} e^{-\frac{1}{2}w^2} dw \quad \because \frac{X^2 - n}{\sqrt{2n}} \text{ is an } N(0,1) \text{ variate}$

Here $\sqrt{2}X^2 - \sqrt{2n-1}$ is approximately a standard normal variate for large n . In other words, $\sqrt{2}X^2$ is an $N(\sqrt{2n-1}, 1)$ variate for large n .

Theorem: If χ_1^2 and χ_2^2 are independent χ^2 -variables with n_1 and n_2 d.f. respectively then $u = \frac{\chi_1^2}{\chi_1^2 + \chi_2^2}$ and $v = \chi_1^2 + \chi_2^2$ are independently distributed, u is a $\beta_1(\frac{n_1}{2}, \frac{n_2}{2})$ variate and v is χ^2 variate with $(n_1 + n_2)$ d.f.

Proof: The joint pdf of χ_1^2 and χ_2^2 is

$$\begin{aligned} dF(\chi_1^2, \chi_2^2) &= f(\chi_1^2) f(\chi_2^2) d\chi_1^2 d\chi_2^2 \quad ; 0 \leq \chi_1^2 < \infty \\ &\quad 0 \leq \chi_2^2 < \infty \\ &= \frac{e^{-\chi_1^2/2} (\chi_1^2)^{\frac{n_1}{2}-1}}{2^{n_1/2} \Gamma(n_1/2)} \cdot \frac{e^{-\chi_2^2/2} (\chi_2^2)^{\frac{n_2}{2}-1}}{2^{n_2/2} \Gamma(n_2/2)} d\chi_1^2 d\chi_2^2 \\ &= \frac{e^{-\frac{\chi_1^2 + \chi_2^2}{2}} (\chi_1^2)^{\frac{n_1}{2}-1} (\chi_2^2)^{\frac{n_2}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} d\chi_1^2 d\chi_2^2 \end{aligned}$$

Since $u = \frac{\chi_1^2}{\chi_1^2 + \chi_2^2}$ and $v = \chi_1^2 + \chi_2^2$

$\Rightarrow \chi_1^2 = uv$, & $\chi_2^2 = v - uv = (1-u)v$

$\therefore J \left(\frac{\chi_1^2, \chi_2^2}{u, v} \right) = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v - uv + uv = v$

$\therefore |J| = v$

$\therefore$ The joint density function of u and v is

$$\begin{aligned} dF(u, v) &= \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} e^{-\frac{v}{2}} (uv)^{\frac{n_1}{2}-1} [(1-u)v]^{\frac{n_2}{2}-1} |J| du dv \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} e^{-\frac{v}{2}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} v^{\frac{n_1+n_2}{2}-2} v du dv \\ &= \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} e^{-\frac{v}{2}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} v^{\frac{n_1+n_2}{2}-1} du dv \\ &\quad ; 0 < u < 1 \\ &\quad 0 < v < \infty \end{aligned}$$

$$\therefore dF(u) = \int_V dF(u,v) dv$$

$$= \int_0^\infty \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{v}{2}} u^{\frac{n_1}{2}-1} v^{\frac{n_1+n_2}{2}-1} (1-u)^{\frac{n_2}{2}-1} du dv$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} du \int_0^\infty e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1} dv$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} du \left(\frac{1}{2}\right)^{-\frac{n_1+n_2}{2}} \sqrt{\frac{n_1+n_2}{2}}$$

$$= \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} du \quad ; 0 < u < 1$$

which is the Beta distribution of first kind with parameters

$\frac{n_1}{2}$ & $\frac{n_2}{2}$.

$$\text{Also, } dF(v) = \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1} dv \int_0^1 u^{\frac{n_1}{2}-1} (1-u)^{\frac{n_2}{2}-1} du$$

$$= \frac{e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1}}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} dv \beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$$

$$= \frac{e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1}}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} dv \frac{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}}{\sqrt{\frac{n_1+n_2}{2}}}$$

$$= \frac{e^{-\frac{v}{2}} v^{\frac{n_1+n_2}{2}-1}}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1+n_2}{2}}} dv \quad ; 0 < v < \infty$$

which is the χ^2 distribution with (n_1+n_2) d.f.

END

Distinguish between sampling distribution and parent distribution.

Solⁿ: (i) The probability distribution of a statistic (sample) is known as sampling distribution.

The distⁿ of population (parameter) is known as parent distribution.

(ii) Sampling distribution is derived from parent distⁿ. Parent distⁿ is not derived from sampling distⁿ.

(iii) χ^2 , t , F -distⁿ are the sampling distⁿ. Normal, Binomial, Poisson distⁿ are the parent distⁿ.

(D.f): The number of independent variates which make up the statistic is known as the degrees of freedom.

Theorem:

Let $x_1, x_2, \dots, x_n$ be a random sample from a normal population with mean μ and variance σ^2 . Then

- (i) $\bar{x} \sim N(\mu, \sigma^2/n)$,
- (ii) $\sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma}\right)^2$ is a χ^2 -variate with $(n-1)$ d.f. and
- (iii) $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\frac{n\sigma^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma}\right)^2$ are independently distributed.

Proof: The joint pdf of $x_1, x_2, \dots, x_n$ is given by

$$dF(x_1, x_2, \dots, x_n) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right] dx_1 dx_2 \dots dx_n$$

$-\infty < (x_1, x_2, \dots, x_n) < \infty$

Let us transform to the variables y_i ($i=1, 2, \dots, n$) by means of a linear orthogonal transformation $Y = AX$.

Let us choose in particular

$$a_{11} = a_{12} = \dots = a_{1n} = 1/\sqrt{n}$$

$$\Rightarrow y_1 = \frac{1}{\sqrt{n}} (x_1 + x_2 + \dots + x_n) = \sqrt{n} \bar{x} \quad \text{--- (1)}$$

Since the transformation is orthogonal, we have

$$\begin{aligned} \sum_{i=1}^n y_i^2 &= \sum_{i=1}^n x_i^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n\bar{x}^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + y_1^2 \quad \text{[from (1)]} \end{aligned}$$

$\sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 = \sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum (x_i - \bar{x})^2$

$$\Rightarrow \sum_{i=2}^n y_i^2 = \sum_{i=1}^n (x_i - \bar{x})^2 \quad \longrightarrow (2)$$

$$\begin{aligned} \sum_{i=1}^n (x_i - \mu)^2 &= \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \\ &= \sum_{i=2}^n y_i^2 + n(\bar{x} - \mu)^2 \quad \text{[from (2)]} \end{aligned}$$

The Jacobian of the transformation is $J = \pm 1$.

Thus the joint density function of $y_1, y_2, \dots, y_n$ is

$$df(y_1, y_2, \dots, y_n) = \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \exp \left[-\frac{1}{2\sigma^2} \left\{ \sum_{i=2}^n y_i^2 + n(\bar{x} - \mu)^2 \right\} \right] \times (J) dy_1 dy_2 \dots dy_n$$

$$= \left[\left(\frac{1}{\sqrt{2\pi} \sigma / \sqrt{n}} \right) \exp \left\{ -\frac{n}{2\sigma^2} (\bar{x} - \mu)^2 \right\} d\bar{x} \right] \times \left[\left(\frac{1}{\sigma \sqrt{2\pi}} \right)^{n-1} \exp \left\{ -\sum_{i=2}^n \frac{y_i^2}{2\sigma^2} \right\} dy_2 dy_3 \dots dy_n \right] \quad (\because dy_1 = \sqrt{n} d\bar{x})$$

Thus $\bar{X}$ and $\sum_{i=2}^n y_i^2 = \sum_{i=1}^n (x_i - \bar{x})^2 = nS^2$ are independently distributed, where moreover $\bar{X} \sim N(\mu, \sigma^2/n)$ and y_i ($i=2, 3, \dots, n$) are independent $N(0, \sigma^2)$. Hence

$$\sum_{i=2}^n \frac{y_i^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^2 = \frac{nS^2}{\sigma^2}, \text{ being the sum of}$$

squares of $(n-1)$ independent standard normal variates is distributed as χ^2 -variate with $(n-1)$ d.f.